LAB 2 ANSWERS REPORT

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Artificial Intelligence (COMP3742)

Table of Contents

Question 1

Load the Iris dataset and display the first five rows. What are the feature names and target classes?

The feature names and their data types are:

- Sepal Length – Continuous
- Sepal Width – Continuous
- Petal Length – Continuous
- Petal Width – Continuous

The target is categorical and the three classes are:

- Iris Setosa
- Iris Versicolour
- Iris Virginica

Question 2

Visualize the pairwise relationships between the features of the Iris dataset. What insights can you draw from the scatter plots?

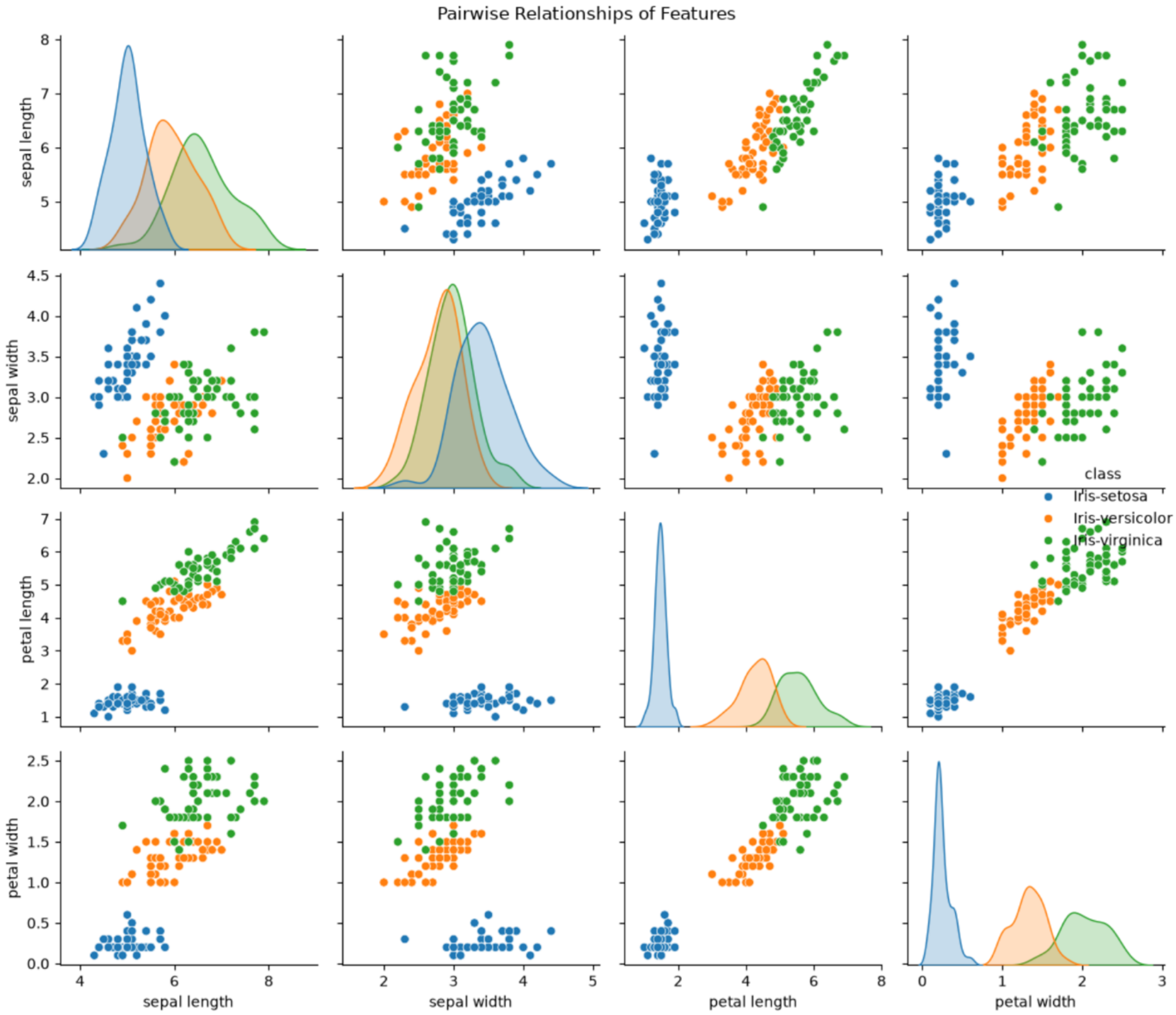

Figure 1 - Pairwise Relations Between Features

Figure 1 - Pairwise Relations Between Features, shows that the Iris Setosa class is very distinct in its associated features which should give the decision tree and easier time partitioning the feature space. The other two seem to be pretty mixed in meaning the tree may need to go deeper to separate these two classes.

Question 3

Train a decision tree classifier on the Iris dataset using scikit-learn. What is the accuracy of this baseline model on the test set?

The baseline accuracy of my model was 1.0, with a test set of 30%. As seen in Figure 2 - DTC Baseline Accuracy.

```
# Train decision tree classifier (no hyperparam tuning)
X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.3, random_state=42)

dtc = DecisionTreeClassifier()
dtc.fit(X_train, y_train)

# Make prediction on test set and print accuracy
dtc_predict = dtc.predict(X_test)

accuracy = accuracy_score(y_test, dtc_predict)
print(f"Prediction Accuracy: {accuracy:.2f}")
✓ [6] 38ms

Prediction Accuracy: 1.00
```

Figure 2 - DTC Baseline Accuracy

I believe this is due to a lucky split since the entire dataset is only 150 instances and test set at 30% is only 45 instances the prediction needed to get correct for 1.0 accuracy.

Question 4

Visualize the decision tree. What are the most important features according to the decision tree?

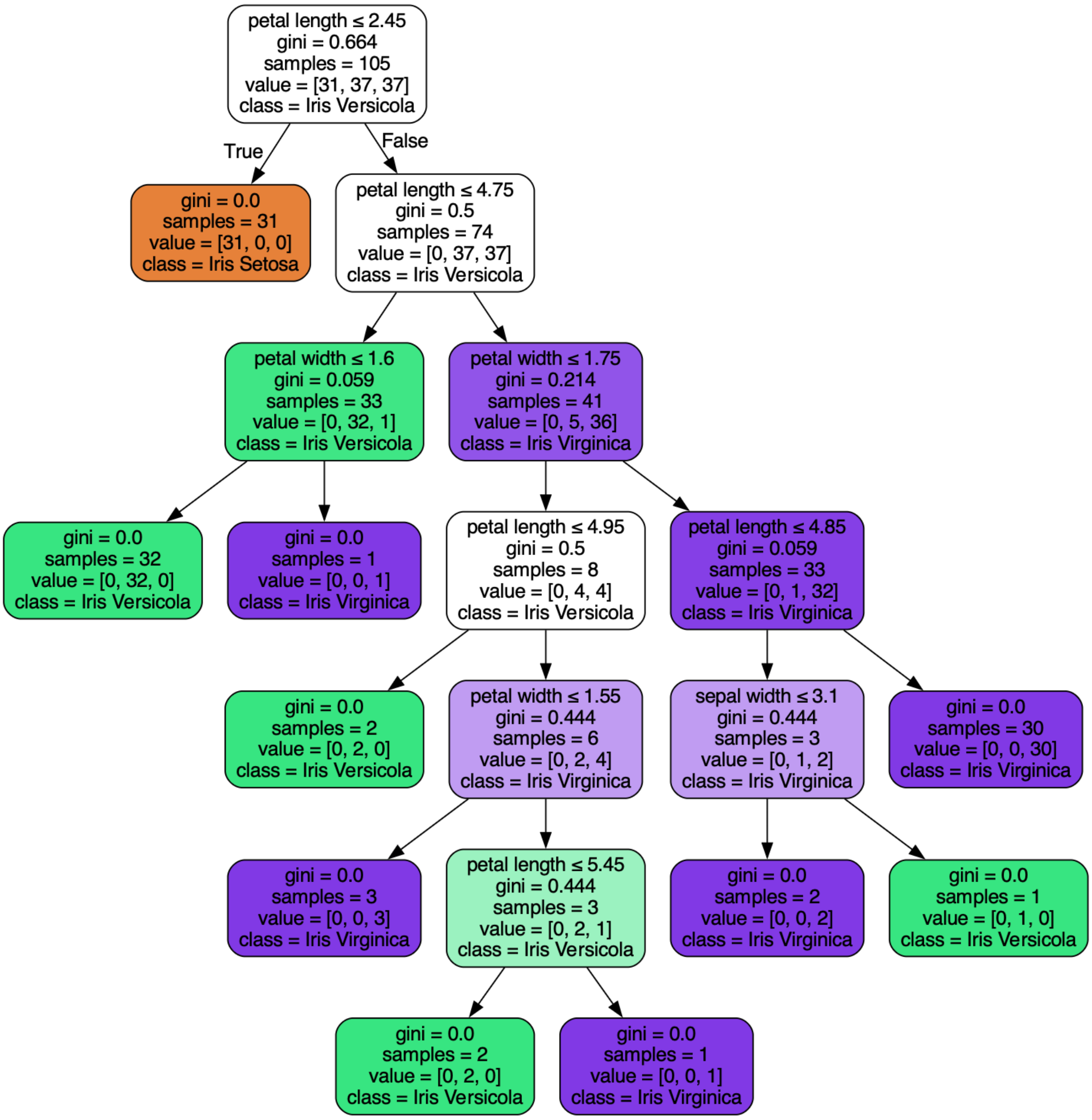

Figure 3 - Decision Tree Visual

As seen in Figure 3 - Decision Tree Visual, petal length and petal width essentially do all the heavy lifting in terms of portioning the feature space making them the most important features by far.

Question 5

Perform hyperparameter optimization for this decision tree classifier using GridSearchCV with 5-fold cross-validation. Identify the best hyperparameters and evaluate the model's performance on the test set. Compare the optimized model's performance with the baseline model.

```python
# Grid search cross validation for finding optimal hyperparameters
hyperparam_grid = {
    'criterion': ['gini', 'entropy', 'log_loss'],
    'max_depth': [None, 10, 20, 30],
    'max_leaf_nodes': [None, 5, 10, 20],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [4, 10, 20, 50],
    'class_weight': [None, 'balanced'],
    'max_features': [None, 'sqrt', 'log2']
}

dtc_gs = DecisionTreeClassifier()
grid_search = GridSearchCV(estimator=dtc_gs, param_grid=hyperparam_grid, cv=10, scoring='accuracy')
grid_search.fit(X_train, y_train)

print("Best Hyper-parameters: ", grid_search.best_params_)
print("Best Score: ", grid_search.best_score_)
```

✓ [8] 1m 12s

```
Best Hyper-parameters:  {'class_weight': None, 'criterion': 'gini', 'max_depth': None, 'max_features': 'sqrt', 'max_leaf_nodes': 20, 'min_samples_leaf': 4, 'min_samples_split': 2}
Best Score:  0.9636363636363636
```

```python
# See test set accuracy
gs_predict = grid_search.predict(X_test)
gs_accuracy = accuracy_score(y_test, gs_predict)

print(f"Grid Search Cross-Validation Accuracy: {gs_accuracy:.2f}")
```

✓ [9] 22ms

```
Grid Search Cross-Validation Accuracy: 0.84
```

Figure 4 - GridSearchCV Accuracy

Figure 4 - GridSearchCV Accuracy shows that we drop down to 0.84 test set accuracy and 0.96 training accuracy. This shows that the grid search forced some non-default parameter values for certain hyperparameters that could have been affecting accuracy on top of the 5 k-fold cross validation ensuring we don't just get a lucky split like on a baseline model.